



AIMS

African Institute for
Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 10: Convergence of the mean, Central Limit Theorem

evans.gouno@univ-ubs.fr

1 The statistic $\overline{X}_n$

2 Central Limit Theorem

WLLN

Weak Law of Large Number (WLLN)

Let $X_1, \dots, X_n$ be n independent and identically distributed (iid) r.v.'s such that $E(X_i) = a < +\infty$ and $\text{Var}(X_i) = b, i = 1, \dots, n$.

Then $\bar{X}_n$ **converge in probability** to a .

Proof: Apply Chebyshev's inequality

WLLN

Weak Law of Large Number (WLLN)

Let $X_1, \dots, X_n$ be n independent and identically distributed (iid) r.v.'s such that $E(X_i) = a < +\infty$ and $\text{Var}(X_i) = b$, $i = 1, \dots, n$.

Then $\bar{X}_n$ **converge in probability** to a .

Proof: Apply Chebyshev's inequality

$$\forall \varepsilon > 0, P(|\bar{X}_n - a| > \varepsilon) \leq \frac{b}{n \varepsilon^2} \xrightarrow{n \rightarrow +\infty} 0$$

$$\iff \forall \varepsilon > 0, \lim_{n \rightarrow +\infty} P(|\bar{X}_n - a| > \varepsilon) = 0$$

$$\iff \bar{X}_n \xrightarrow{P} a.$$

Khinchin's theorem

Khinchin's theorem

Let $X_1, \dots, X_n$ be n independent and identically distributed (iid) r.v.'s such that $E(X_i) = a < +\infty$, $i = 1, \dots, n$.

Then $\bar{X}_n$ is a **consistent unbiased estimator** of a .

Khinchin's theorem

Khinchin's theorem

Let $X_1, \dots, X_n$ be n independent and identically distributed (iid) r.v.'s such that $E(X_i) = a < +\infty$, $i = 1, \dots, n$.

Then $\bar{X}_n$ is a **consistent unbiased estimator** of a .

Proof:

Khinchin's theorem

Khinchin's theorem

Let $X_1, \dots, X_n$ be n independent and identically distributed (iid) r.v.'s such that $E(X_i) = a < +\infty$, $i = 1, \dots, n$.

Then $\bar{X}_n$ is a **consistent unbiased estimator** of a .

Proof:

- Convergence of m.g.f: $M_{\bar{X}_n}(t) \xrightarrow{n \rightarrow +\infty} e^{ta}$.

Khinchin's theorem

Khinchin's theorem

Let $X_1, \dots, X_n$ be n independent and identically distributed (iid) r.v.'s such that $E(X_i) = a < +\infty$, $i = 1, \dots, n$.

Then $\bar{X}_n$ is a **consistent unbiased estimator** of a .

Proof:

- Convergence of m.g.f: $M_{\bar{X}_n}(t) \xrightarrow{n \rightarrow +\infty} e^{ta}$.
- $\implies$ Convergence in distribution: $\bar{X}_n \xrightarrow{d} X$,
where $X \sim \delta_a$, the **point mass** (or **Dirac**) **distribution**.

Khinchin's theorem

Khinchin's theorem

Let $X_1, \dots, X_n$ be n independent and identically distributed (iid) r.v.'s such that $E(X_i) = a < +\infty$, $i = 1, \dots, n$.

Then $\bar{X}_n$ is a **consistent unbiased estimator** of a .

Proof:

- Convergence of m.g.f: $M_{\bar{X}_n}(t) \xrightarrow{n \rightarrow +\infty} e^{ta}$.
- $\implies$ Convergence in distribution: $\bar{X}_n \xrightarrow{d} X$,
where $X \sim \delta_a$, the **point mass** (or **Dirac**) **distribution**.
- $\implies$ Convergence in Probability: $\implies \bar{X}_n \xrightarrow{P} a$.

Limit of $\bar{X}_n$ m.g.f

Proof of the first point:

Limit of $\bar{X}_n$ m.g.f

Proof of the first point:

$$M_{\bar{X}_n}(t) = E(e^{t\bar{X}_n}) = E\left(e^{\frac{t}{n} \sum_{i=1}^n X_i}\right) = \prod_{i=1}^n E\left(e^{\frac{t}{n} X_i}\right) = \left[E\left(e^{\frac{t}{n} X_i}\right)\right]^n$$

$$\text{But } e^{\frac{t}{n} X_i} = 1 + \frac{t}{n} X_i + \frac{t^2}{n^2} X_i^2 + \dots$$

$$\text{Then } E\left(e^{\frac{t}{n} X_i}\right) \sim \left[1 + \frac{t}{n} E(X_i) + o\left(\frac{t}{n}\right)\right]^n = \left[1 + \frac{ta}{n} + o\left(\frac{t}{n}\right)\right]^n$$

where o is a function such that $\lim_{h \rightarrow 0} o(h)/h = 0$.

$$\begin{aligned} \left[1 + \frac{ta}{n} + o\left(\frac{t}{n}\right)\right]^n &= \exp\left\{n \log\left[1 + \frac{at + n o(t/n)}{n}\right]\right\} \\ &= \exp\left\{n \times \frac{at + n o(t/n)}{n}\right\} \\ &= \exp\{at + n o(t/n)\} \end{aligned}$$

which goes to $\exp\{at\}$ when n goes to infinity since $n o(t/n)$ goes to 0.

Hence

$$M_{\bar{X}_n}(t) \xrightarrow{n \rightarrow +\infty} e^{ta}.$$

Convergence in distribution

$$\implies \bar{\mathbf{X}}_n \xrightarrow{d} \mathbf{X}, \text{ where } X \sim \delta_a.$$

Convergence in distribution

Z_n converges **in distribution** or **in law** if,

$$\lim_{n \rightarrow +\infty} F_{Z_n}(z) = F_Z(z),$$

at each point z where F_Z is continuous.

Notation: $Z_n \xrightarrow{d} Z$ or $Z_n \xrightarrow{\mathcal{L}} Z$

Point mass distribution

$$\Rightarrow \bar{X}_n \xrightarrow{d} X, \text{ where } \mathbf{X} \sim \delta_{\mathbf{a}}.$$

Point mass distribution

A r.v. X on $\mathbb{R}$, is said to follow the **point mass distribution** δ_c , if $P(X = c) = 1$.

The c.d.f of X is defined as:

$$F(x) = \begin{cases} 0 & x < c \\ 1 & x \geq c \end{cases}$$

Moment generating function:

$$M_X(t) = E(e^{tX}) = e^{tc} P(X = c) = e^{tc}.$$

Convergence in distribution $\overline{X}_n$

Continuity Theorem for m.g.f.'s

Let $Z_1, Z_2, \dots$ be a sequence of r.v.'s, F_n be the c.d.f of Z_n and $M_{Z_n}(t)$ the m.g.f of Z_n . Then:

$$\left(M_{Z_n}(t) \xrightarrow{n \rightarrow +\infty} M_Z(t) \right) \iff \left(Z_n \xrightarrow{d} Z \right).$$

Proposition

Let $X_1, X_2, \dots$ be a sequence of r.v.'s, with F_n , the c.d.f of X_n . Let X be a r.v with the point mass distribution δ_c . Then:

$$\left(X_n \xrightarrow{d} X \right) \iff \left(X_n \xrightarrow{P} c \right).$$

Proof of the proposition

- Suppose $\lim_{n \rightarrow +\infty} F_{X_n}(x) = F_X(x)$ with $X \sim \delta_c$.

$$\begin{aligned} \forall \varepsilon > 0, P(|X_n - c| > \varepsilon) &= P(X_n > c + \varepsilon) + P(X_n < c - \varepsilon) \\ &= 1 - F_n(c + \varepsilon) + F_n(c - \varepsilon) \end{aligned}$$

$$\lim_{n \rightarrow +\infty} F_{X_n}(c - \varepsilon) = F_X(c - \varepsilon) = 0 \text{ and } \lim_{n \rightarrow +\infty} F_{X_n}(c + \varepsilon) = F_X(c + \varepsilon) = 1.$$

Hence $X_n \xrightarrow{P} c$.

- Suppose that $\forall \varepsilon > 0, P(|X_n - c| > \varepsilon) \rightarrow 0$ when $n \rightarrow +\infty$.

Then $\lim_{n \rightarrow +\infty} [P(X_n > c + \varepsilon) + P(X_n < c - \varepsilon)] \rightarrow 0$ which means that

$$\lim_{n \rightarrow +\infty} P(X_n > c + \varepsilon) = 0 \text{ and } \lim_{n \rightarrow +\infty} P(X_n < c - \varepsilon) = 0.$$

Therefore

$$\forall \varepsilon > 0, \lim_{n \rightarrow +\infty} F_{X_n}(c - \varepsilon) = 0 \text{ and } \lim_{n \rightarrow +\infty} F_{X_n}(c + \varepsilon) = 1.$$

Hence $\lim F_n(x) = F_X(x)$ such that $F_X(x) = 1$ for $x \geq c$ and $F_X(x) = 0$ for $x < c$ which means that X_n converges to X having a point mass probability δ_c .

CLT

Central Limit Theorem (CLT)

Let $X_1, \dots, X_n$ be n independent and identically distributed (iid) r.v.'s with $E(X_1) = \mu$ and $Var(X_1) = \sigma^2 > 0$ (both finite).

Then:

$$\sqrt{n} \left(\frac{\bar{X}_n - \mu}{\sigma} \right) \sim \mathcal{N}(0, 1) \text{ when } n \rightarrow +\infty.$$

The r.v. $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$ converges **in distribution** (or in law) to a standard normal r.v. or X_n behaves as a normal distribution with parameters $(\mu, \sigma^2/n)$ when n goes to infinity.

$$\lim_{n \rightarrow +\infty} F_{Z_n}(z) = \Phi(z)$$

CLT

Proof: We are going to show that $M_{Z_n}(t) \xrightarrow{n \rightarrow +\infty} M_Z(t)$.

$$Z_n = \sqrt{n} \left(\frac{\bar{X}_n - \mu}{\sigma} \right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)$$

$$\text{Let } Y_i = \frac{X_i - \mu}{\sigma}, \quad Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i.$$

$$M_{Z_n}(t) = E(e^{tZ_n}) = \prod_{i=1}^n E(e^{\frac{t}{\sqrt{n}} Y_i}) = \left[E(e^{\frac{t}{\sqrt{n}} Y_i}) \right]^n.$$

$$e^{\frac{t}{\sqrt{n}} Y_i} \approx 1 + \frac{t}{\sqrt{n}} Y_i + \frac{t^2}{2n} Y_i^2 \iff E(e^{\frac{t}{\sqrt{n}} Y_i}) \approx 1 + \frac{t}{\sqrt{n}} E(Y_i) + \frac{t^2}{2n} E(Y_i^2)$$

Then $E(e^{\frac{t}{\sqrt{n}} Y_i}) \approx 1 + \frac{t^2}{2n}$ since $E(Y_i) = 0$ and $\text{Var}(Y_i) = E(Y_i^2) = 1$.

$$\left[E(e^{\frac{t}{\sqrt{n}} Y_i}) \right]^n \approx \left(1 + \frac{t^2}{2n} \right)^n.$$

But $\left(1 + \frac{t^2}{2n} \right)^n = e^{n \log(1 + \frac{t^2}{2n})} \approx e^{\frac{t^2}{2}}$ since $\log(1 + \frac{t^2}{2n}) \approx \frac{t^2}{2n}$ when n large.

Therefore $\lim_{n \rightarrow +\infty} M_{Z_n}(t) = e^{\frac{t^2}{2}}$ which is the m.g.f. of a standard normal distribution.

Conclusion: $Z_n \xrightarrow{d} Z$, where $Z \sim \mathcal{N}(0, 1)$.

CLT

Example:

Let $X_1, \dots, X_n$ be independent Bernoulli r.v.'s with parameter p .

We have $E(X_i) = p$ and $\text{Var}(X_i) = p(1 - p)$.

Applying the CLT leads to :

$$\sqrt{n} \left(\frac{\bar{X}_n - p}{\sqrt{p(1 - p)}} \right) \sim \mathcal{N}(0, 1).$$

Simulation:

- Choose a value for p e.g. $p = 0.8$.
- Draw 100 realizations of a Bernoulli with parameter 0.8.
- Compute $\bar{X}_n$, the mean of the 100 values.
- Reproduce this experiment 1000 times to obtain a sample of $\bar{X}_n$.
- Build the histogram to investigate the distribution of the variables

$$\sqrt{n}((\bar{X}_n - p)/\sqrt{p(1 - p)}).$$

CLT for Bernoulli



R code

```
p<-0.8 # Choose a value for p
n<-100 # Size of the sample
# Generate a sample of n Bernoulli r.v.'s realizations
x<-sample(c(0,1), n, replace=TRUE, prob=c(0.2,0.8))
M<-1000 # Number of samples
#  $Z = \sqrt{n}(\bar{X}_n - p) / \sqrt{p(1-p)}$ 
Z<-rep(0,M) # Initialization
for (i in 1:M)
Z[i]<-sqrt(n)*((mean(rbinom(n,1,p))-p)/sqrt(p*(1-p)))
# Histogram
hist(Z, nclass=20, probability=T, main=list("CLT for
Bernoulli",cex=1.5),
ylab="Density", yaxt = "n")
# Adding the p.d.f.
z<-seq(-3,3,0.01)
gaussz<-dnorm(z,0,1)
lines(z,gaussz)
box() # To add a frame
```